

CSE 4255: Telecommunication

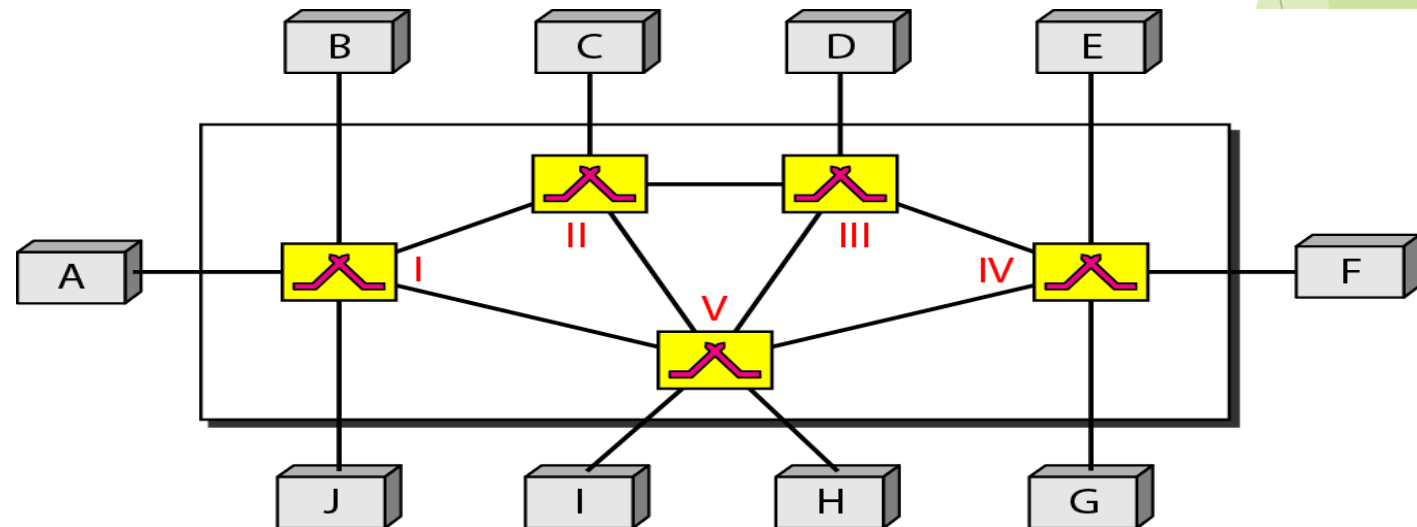
Lecture 3: Switching

Why Switching?

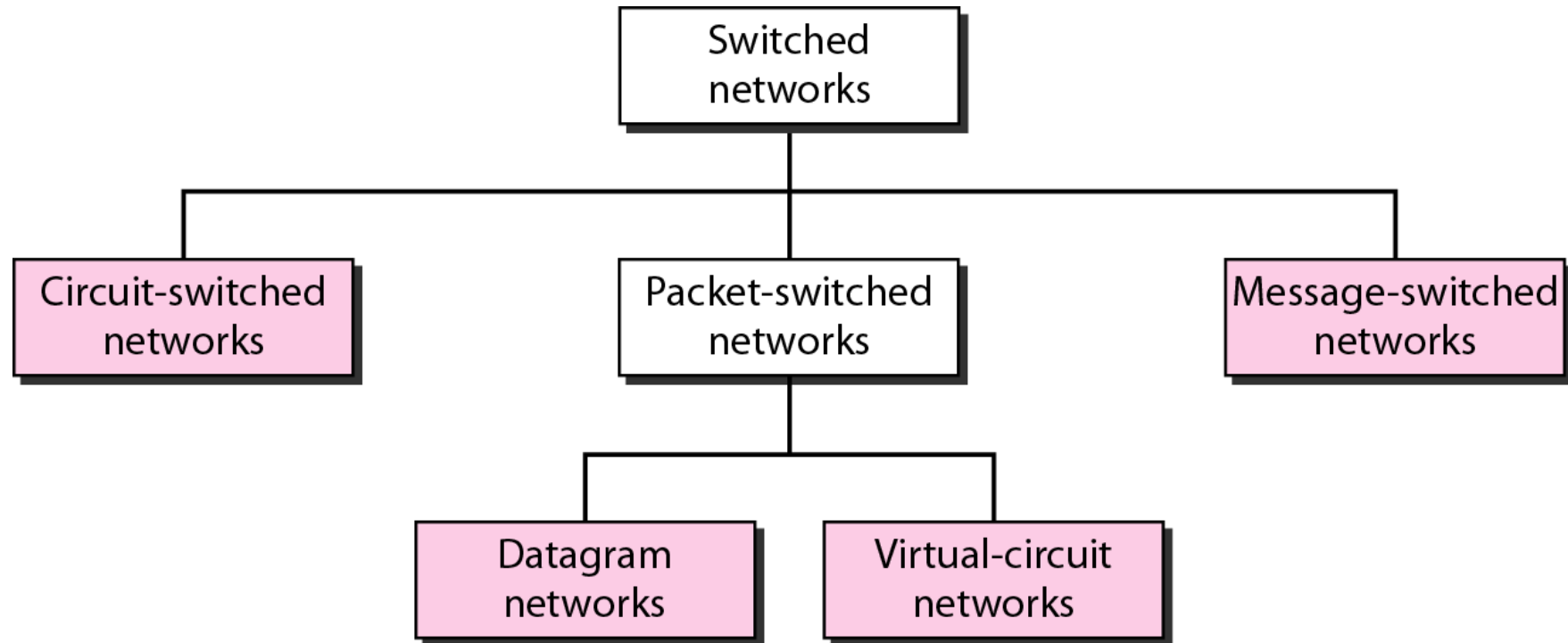
- ▶ A network is a set of connected devices. Some of the conventional ways of interconnecting devices are:
 - a) Point-to point connection between devices as in a mesh topology.
 - b) Connection between a central device and every other device as in star topology.
 - c) Bus topology – not practical if the devices are at great distances.
- All these techniques require extensive cabling, dependence on a central server or a central bus.
- The solution to this interconnectivity problem is switching.

Switching Network:

- ▶ A switched network consists of a series of interlinked nodes, called switches.
- ▶ Switches are devices capable of creating temporary connections between two or more devices linked to the switch.
- ▶ In a switched network, some of these nodes are connected to the end systems (computers or telephones, for example). Others are used only for routing.

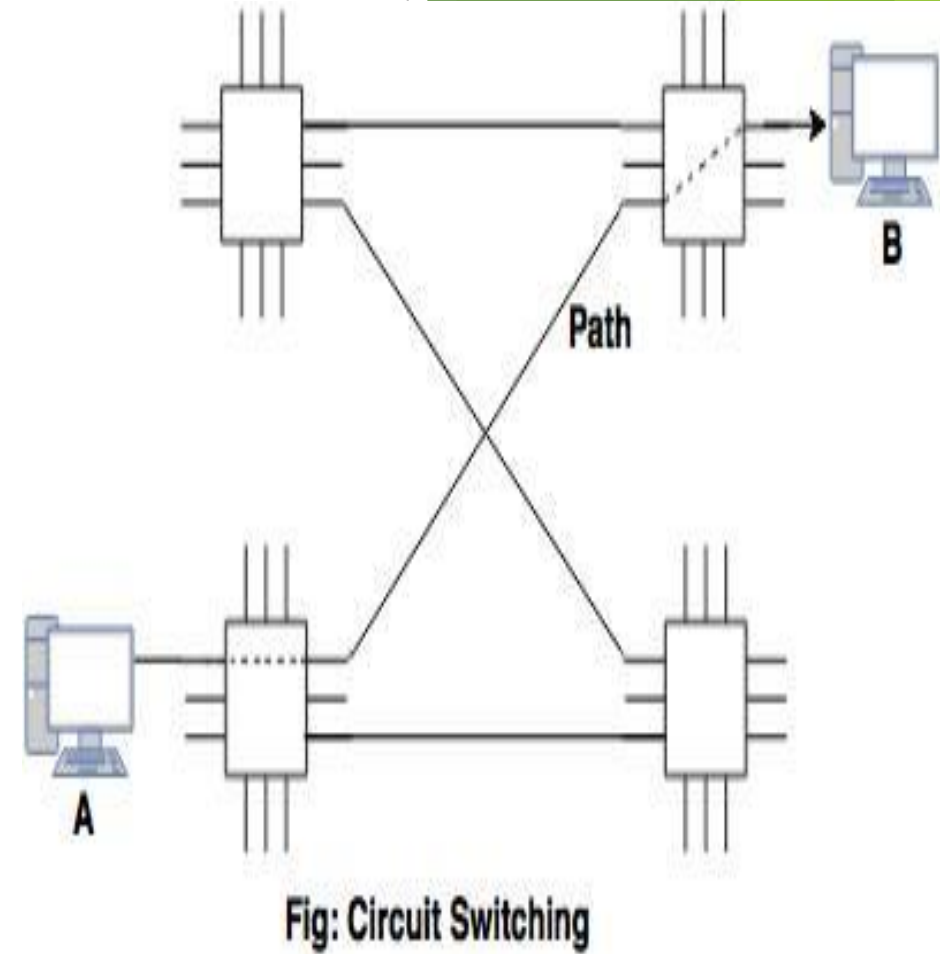


Types of Switching Network:



Circuit Switching

- ▶ Circuit switching takes place at the physical layer.
- ▶ The actual communication in a circuit-switched network requires three phases: connection setup, data transfer, and connection teardown.
- ▶ Before starting communication, the stations must make a reservation for the resources to be used during the communication.
- ▶ Data transferred between the two stations are a continuous flow.
- ▶ There is no addressing involved during data transfer but there is end-to-end addressing used during the setup phase.



Message Switching

- ▶ With message switching there is no need to establish a dedicated path between two stations.
- ▶ When a station sends a message, the destination address is appended to the message.
- ▶ Each node receives the entire message, stores it in its buffer, and then transmits the message to the next node.
- ▶ This type of network is called a store-and-forward network.

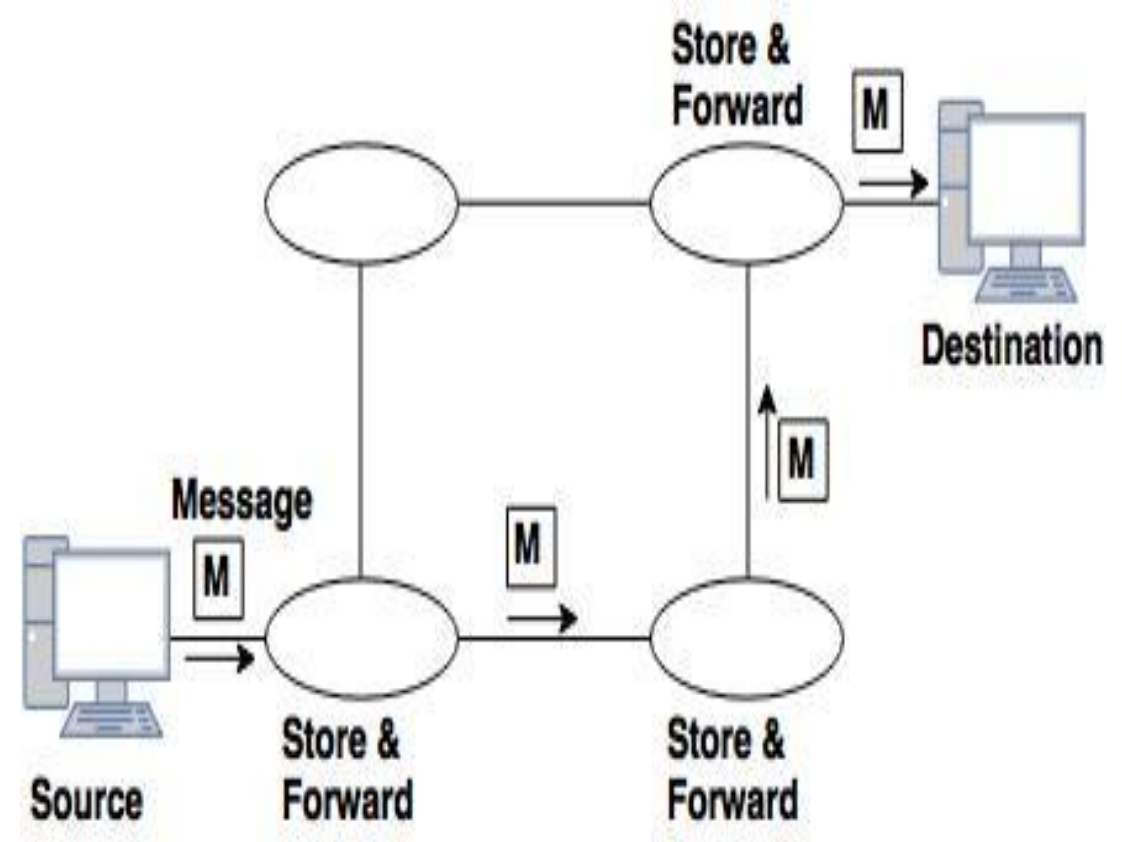


Fig: Message Switching

Packet Switching

- ▶ There are two methods of packet switching: Datagram and virtual circuit.
- ▶ In both packet switching methods, a message is broken into small parts, called packets.
- ▶ Each packet is tagged with appropriate source and destination addresses.
- ▶ With current technology, packets are generally accepted onto the network on a first-come, first-served basis. If the network becomes overloaded, packets are delayed or discarded ("dropped").

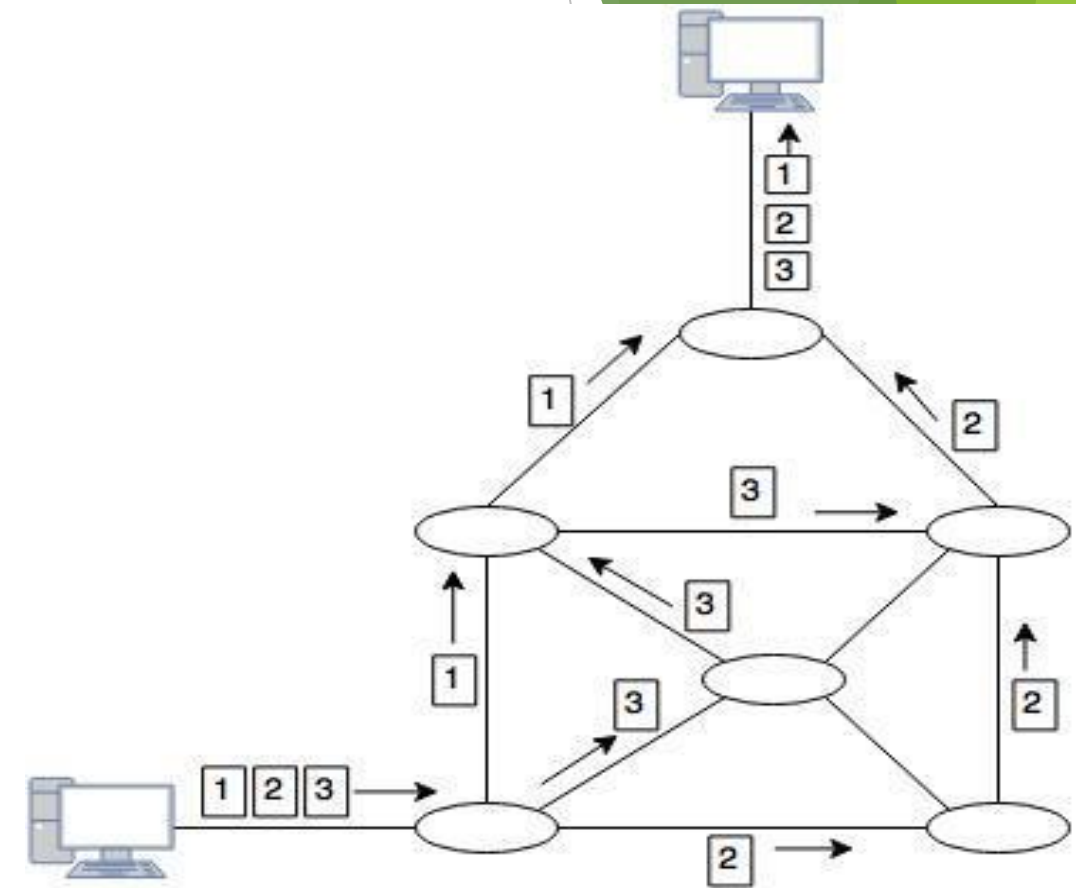


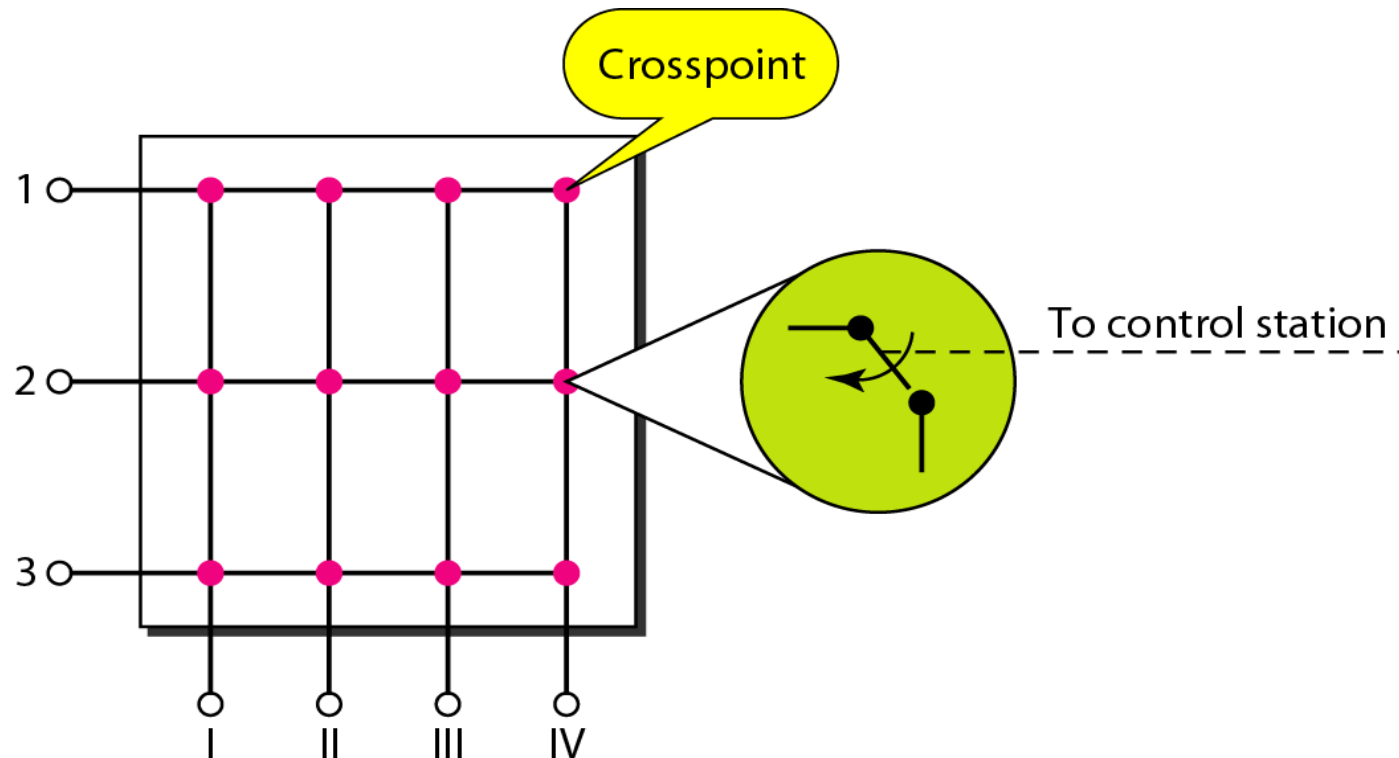
Fig: Datagram Packet Switching

Circuit Switching (Space-division switch)

- ▶ In space division switching, the paths in the circuit are separated with each other spatially.
- ▶ Different ongoing connections, at a same instant of time, uses different switching paths, which are separated spatially.
- ▶ Some of the space switches are
 1. Crossbar switch
 2. Multi-stage switch

Crossbar Switch

- ▶ A crossbar switch connects n inputs to m outputs in a grid, using electronic microswitches (transistors) at each crosspoint.
- ▶ As all the stations are allowed to be connected with all possible connections as long as the called party is free, this Crossbar Switching is called the Non-Blocking.



Limitations

- ▶ The major limitation of this design is the number of crosspoints required.
- ▶ To connect n inputs to m outputs using a crossbar switch requires $n \times m$ crosspoints.
- ▶ For example, to connect 1000 inputs to 1000 outputs requires a switch with 1,000,000 crosspoints.
- ▶ A crossbar with this number of crosspoints is impractical.
- ▶ Such a switch is also inefficient because statistics show that, in practice, fewer than 25 percent of the crosspoints are in use at any given time. The rest are idle.

Multistage Switch-Why?

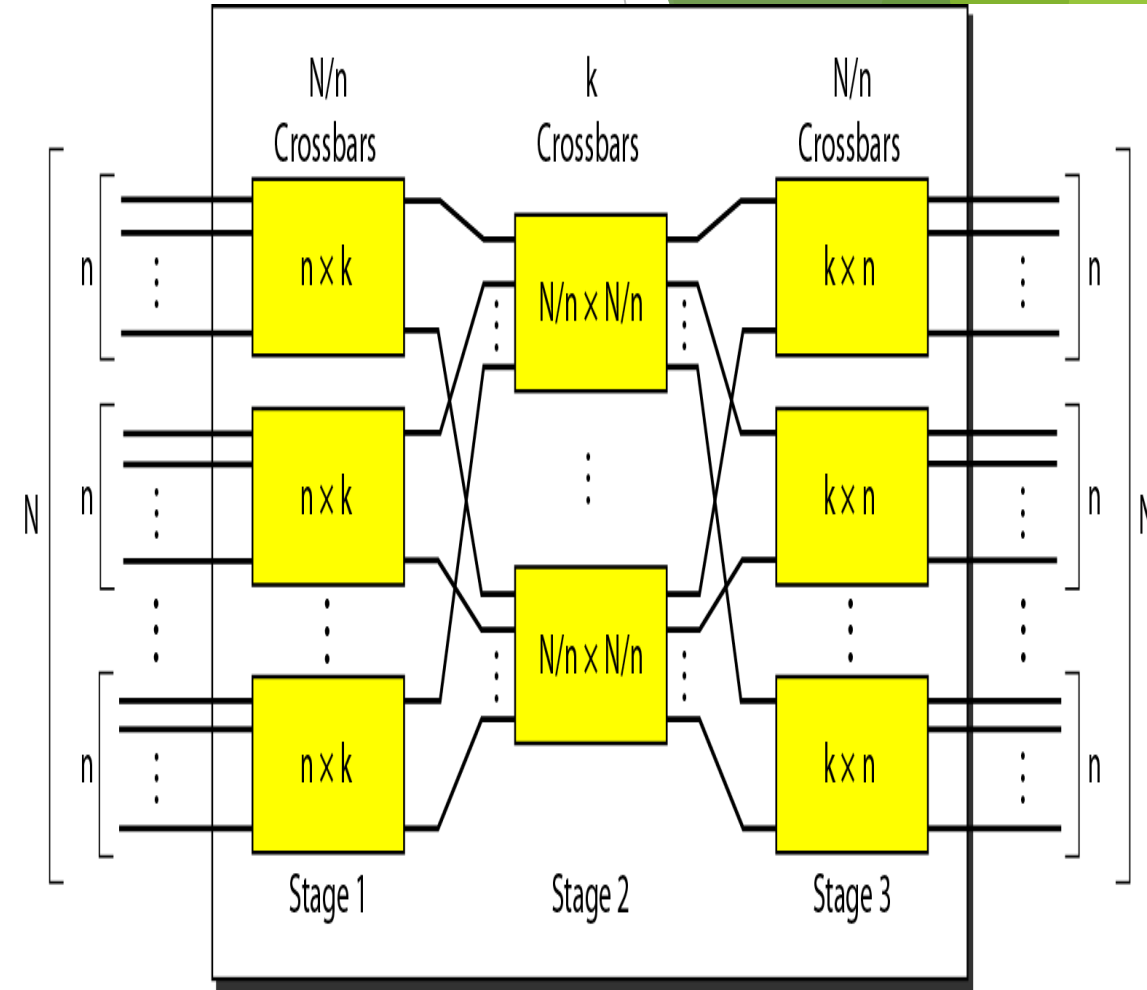
- ▶ The solution to the limitations of the crossbar switch is the multistage switch, which combines crossbar switches in several (normally three) stages.
- ▶ In a single crossbar switch, only one row or column (one path) is active for any connection. So we need $N \times N$ crosspoints.
- ▶ If we can allow multiple paths inside the switch, we can decrease the number of crosspoints.
- ▶ Each crosspoint in the middle stage can be accessed by multiple crosspoints in the first or third stage.

Multistage Switch Structure

► To design a three-stage switch, we follow these steps:

1. We divide the N input lines into groups, each of n lines. For each group, we use one crossbar of size $n \times k$, where k is the number of crossbars in the middle stage. In other words, the first stage has N/n crossbars of $n \times k$ crosspoints.
2. We use k crossbars, each of size $(N/n) \times (N/n)$ in the middle stage.
3. We use N/n crossbars, each of size $k \times n$ at the third stage.

► We can calculate the total number of crosspoints as follows:



$$N/n (n \times k) + k (N/n \times N/n) + N/n (k \times n) = 2kN + k(N/n)^2$$

Example: Multistage Switch

Design a three-stage, 200×200 switch ($N = 200$) with $k = 4$ and $n = 20$.

Solution

In the *first stage* we have N/n or 10 crossbars,
each of size 20×4 .

In the *second stage*, we have 4 crossbars,
each of size 10×10 .

In the *third stage*, we have 10 crossbars,
each of size 4×20 .

The total number of crosspoints is $2kN + k(N/n)^2$, or 2000 crosspoints.

This is 5 percent of the number of crosspoints in a single-stage switch
($200 \times 200 = 40,000$).

Blocking or Non-blocking

► Blocking

- Blocking refers to times when one input cannot be connected to an output because there is no path available between them-all the possible intermediate switches are occupied.
- In a single-stage switch, blocking does not occur because every combination of input and output has its own crosspoint; there is always a path.
- In a multistage switch, The small number of crossbars at the middle stage creates blocking; if $k < n$, the switch is blocked.

► Non-blocking

- Non-blocking permits all stations to connect (in pairs) at once.
- Clos investigated the condition of nonblocking in multistage switches and came up with the following formula.
- According to the Clos criterion: $n = (N/2)^{1/2}$

$$k \geq 2n - 1$$

Example- Non-blocking

Redesign the previous three-stage, 200×200 switch, using the Clos criteria with a minimum number of crosspoints.

Solution

We let $n = (200/2)^{1/2}$, or $n = 10$.

We calculate $k = 2n - 1 = 19$.

In the *first stage*, we have $200/10$, or 20, crossbars,
each with 10×19 crosspoints.

In the *second stage*, we have 19 crossbars,
each with 20×20 crosspoints.

In the *third stage*, we have 20 crossbars
each with 19×10 crosspoints.

The total number of crosspoints is

$$20(10 \times 19) + 19(20 \times 20) + 20(19 \times 10) = 15200.$$